



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

COMP 5212

Machine Learning

Instructor: Zihan Zhang

Lecture 4: Linear Regression

Learning Goals

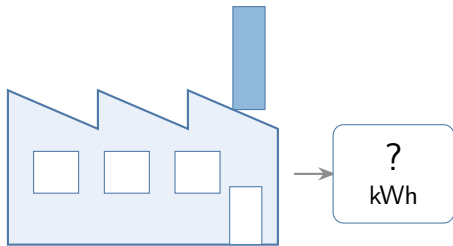
By the end of this lecture, you should be able to:

- formulate linear regression and justify the squared-error loss;
- derive and apply the ordinary least squares (OLS) solution;
- explain and compare Ridge and Lasso regularization;
- use polynomial features and kernels to model nonlinear relationships;
- interpret an OLS generalization bound and explain its assumptions.

Motivation Example

Tomorrow's electricity use?

A factory plans to produce 5.5 tonnes tomorrow.

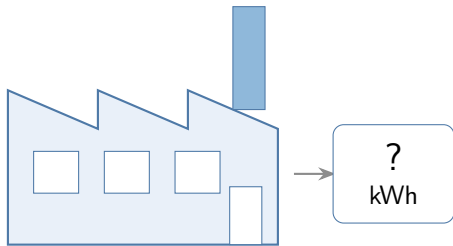


Production plan

What would you measure?

Tomorrow's electricity use?

A factory plans to produce 5.5 tonnes tomorrow.



Production plan

Past daily records

Production + electricity

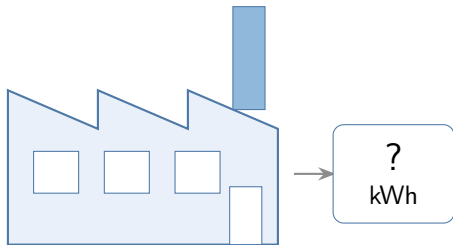
Weather + shifts

Tomorrow's target

Electricity use

Tomorrow's electricity use?

A factory plans to produce 5.5 tonnes tomorrow.



Inputs x

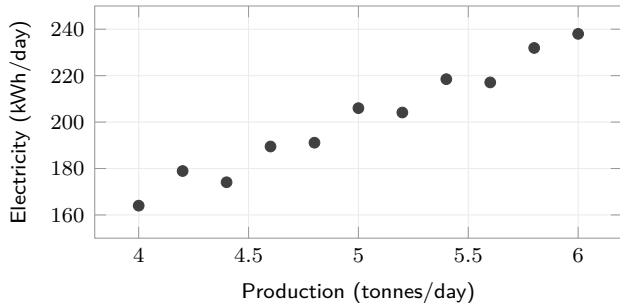


Prediction $\hat{y}$

Budgeting Energy monitoring

Tomorrow's electricity use?

Tomorrow: 5.5 tonnes. Your forecast?

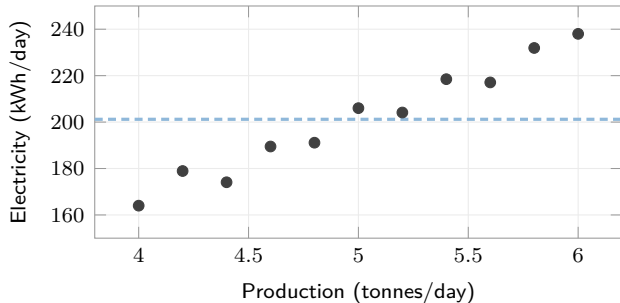


One number?

Explain your choice.

Tomorrow's electricity use?

Tomorrow: 5.5 tonnes. Your forecast?



Ignore production

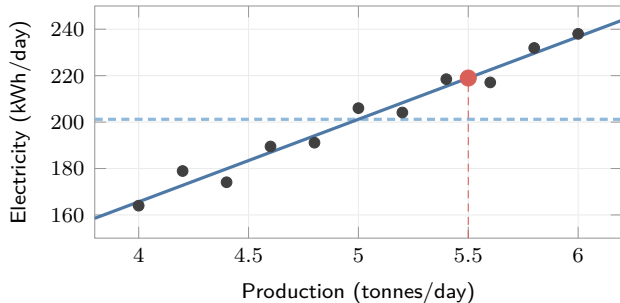
$$\bar{y} = 201.2$$

kWh/day

Where does this miss?

Tomorrow's electricity use?

Tomorrow: 5.5 tonnes. Your forecast?



Use production

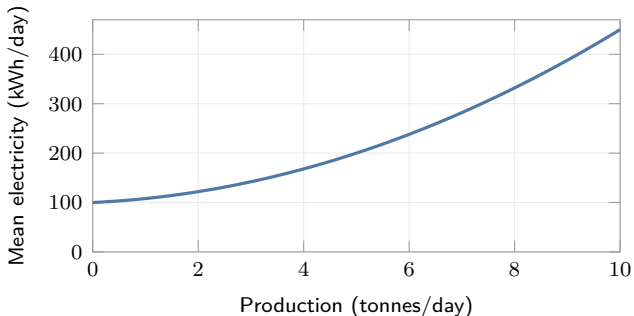
$$\hat{y} = b + wx$$

$$\hat{y}(5.5) \approx 219$$

kWh/day

But the world is nonlinear

Would one straight line work everywhere?

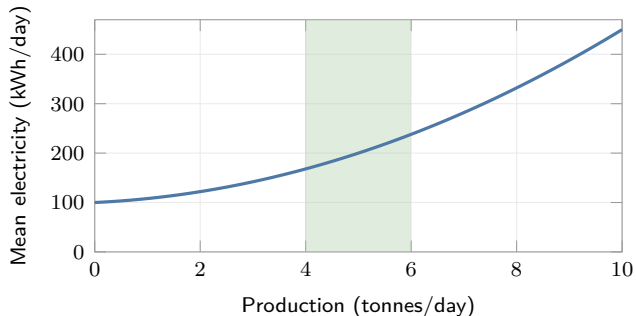


Suppose the true mean looks like this.

Reject a linear model?

But the world is nonlinear

Would one straight line work everywhere?



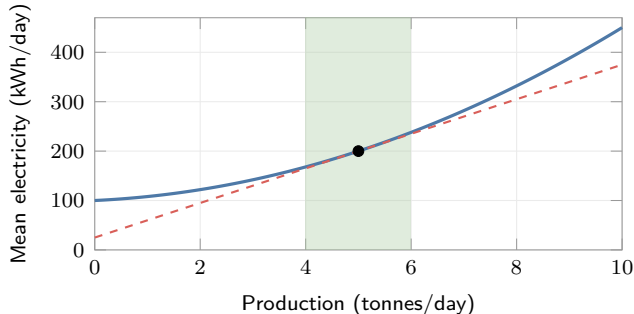
Where do we operate?

Observed range:

$$4 \leq x \leq 6$$

But the world is nonlinear

Would one straight line work everywhere?



Curved mean

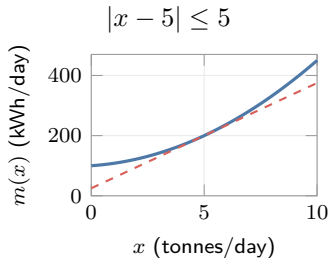
Local tangent

Close here.

Different elsewhere.

Zoom in. What changes?

Same curve. Same tangent. Smaller neighborhoods.



Max. gap: 75 kWh

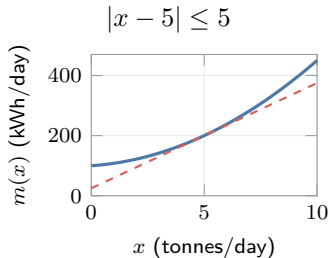
Solid: mean

Dashed: tangent

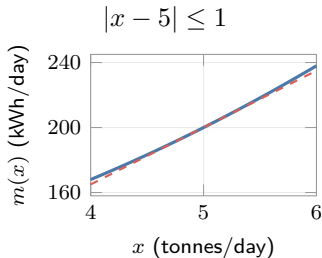
Each panel rescales its axes.

Zoom in. What changes?

Same curve. Same tangent. Smaller neighborhoods.



Max. gap: 75 kWh



Max. gap: 3 kWh

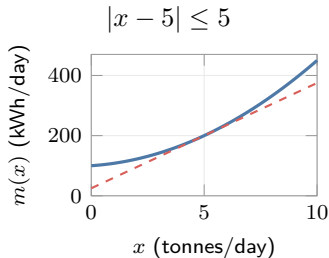
Solid: mean

Dashed: tangent

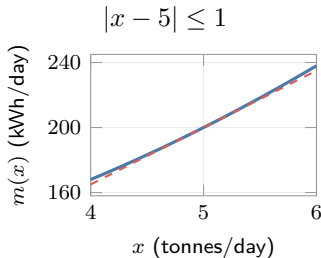
Each panel rescales its axes.

Zoom in. What changes?

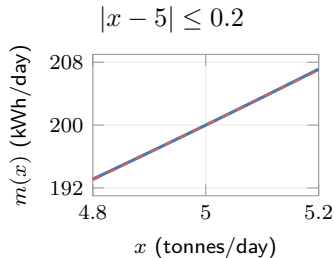
Same curve. Same tangent. Smaller neighborhoods.



Max. gap: 75 kWh



Max. gap: 3 kWh



Max. gap: 0.12 kWh

Solid: mean

Dashed: tangent

Each panel rescales its axes.

Useful range depends on curvature, noise, and required accuracy.

Why linear models appear everywhere

What is the first change near an operating point?

$$x_0 \longrightarrow x_0 + h$$

What happens to $m(x)$?

Why linear models appear everywhere

What is the first change near an operating point?

$$\underbrace{m(x_0 + h)}_{\text{nearby mean}} \approx \underbrace{m(x_0)}_{\text{level}} + \underbrace{m'(x_0)}_{\text{slope}} h.$$

Level + sensitivity $\times$ change

Why linear models appear everywhere

What is the first change near an operating point?

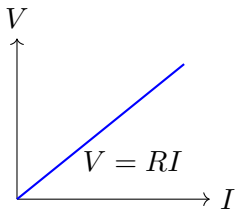
$$\begin{aligned}m(x_0 + h) &\approx m(x_0) + \nabla m(x_0)^\top h, \\ &= b + w^\top x, \quad x = x_0 + h.\end{aligned}$$

$$w = \nabla m(x_0), \quad b = m(x_0) - w^\top x_0$$

Why linear models appear everywhere

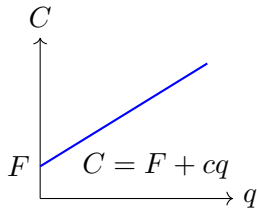
Linear relationships arise naturally in many settings.

Physical laws



Ohm's law: voltage is proportional to current for a fixed resistance.

Economic models



Total cost is linear in output when marginal cost is constant.

How much complexity do we need?

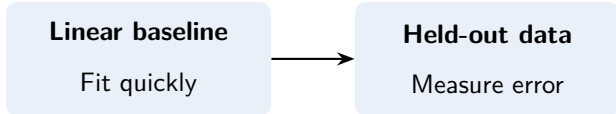
Unknown curvature. Limited observations. A new prediction task.

Straight line? Polynomial? Neural network?

What would you try first?

How much complexity do we need?

Unknown curvature. Limited observations. A new prediction task.

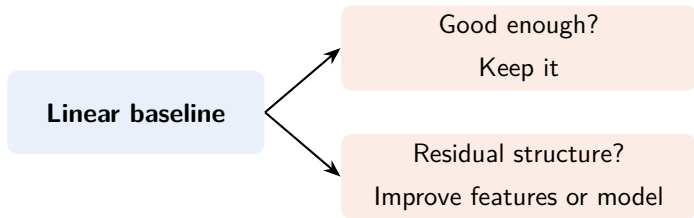


A first benchmark, before adding complexity.

An inexpensive starting point; adequacy is an empirical question.

How much complexity do we need?

Unknown curvature. Limited observations. A new prediction task.



An inexpensive starting point; adequacy is an empirical question.

Formulation and closed-form derivation

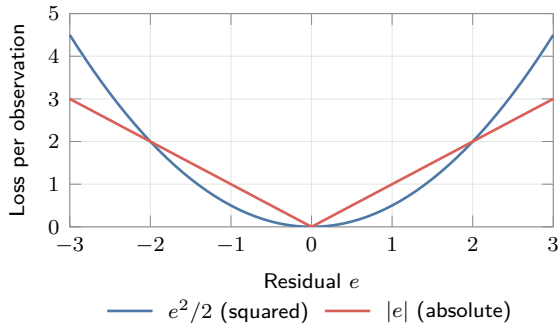
Linear in the parameters

$$f_{\theta}(x) = b + w^{\top}x \quad \theta = [b, w]^{\top}$$

- Weights enter linearly in input.
- The intercept b is the prediction when every feature is zero.

What counts as a good fit?

Squared loss gives larger errors more weight



Residual: $e_i = y_i - f_\theta(x_i)$.

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n e_i^2.$$

- $1/2$ simplifies derivatives.
- $1/n$ normalizes sample size.

Build the design matrix first

Rows represent observations; columns represent the features whose effects we fit

$$X = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n,1} & \cdots & x_{n,d} \end{bmatrix}, \quad \theta = \begin{bmatrix} b \\ w \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

- With $p = d + 1$, $X \in \mathbb{R}^{n \times p}$, $\theta \in \mathbb{R}^p$, and $y \in \mathbb{R}^n$; assume $n > 0$.
- The i th row is $\tilde{x}_i^\top$ (where $\tilde{x} = [1, x^\top]^\top$), so $(X\theta)_i = \theta^\top \tilde{x}_i$.
- Predictions and residuals are $\hat{y} = X\theta$ and $e = y - X\theta$.

Expand the quadratic before differentiating

Expose the terms that depend on the parameter vector

$$\begin{aligned}2nJ(\theta) &= (X\theta - y)^\top (X\theta - y), \\&= \theta^\top X^\top X\theta - \theta^\top X^\top y - y^\top X\theta + y^\top y, \\y^\top X\theta &= (y^\top X\theta)^\top = \theta^\top X^\top y, \\2nJ(\theta) &= \theta^\top X^\top X\theta - 2\theta^\top X^\top y + y^\top y.\end{aligned}$$

- Both cross terms are the same scalar; a scalar equals its transpose.
- $y^\top y$ is constant in θ , while $X^\top X$ is symmetric.

Derive the gradient using a differential

A small parameter change gives $dr = X d\theta$ for $r = X\theta - y$

$$\begin{aligned}dJ &= \frac{1}{2n} [(dr)^\top r + r^\top dr], \\&= \frac{1}{n} r^\top X d\theta = \left[\frac{1}{n} X^\top (X\theta - y) \right]^\top d\theta, \\ \nabla J(\theta) &= \frac{1}{n} X^\top (X\theta - y).\end{aligned}$$

- Match the coefficient of $d\theta$ with $dJ = (\nabla J)^\top d\theta$.
- Component j : $\partial J / \partial \theta_j = n^{-1} \sum_i X_{ij} (X\theta - y)_i$.

Normal equations give a global minimum

A zero gradient is sufficient here because the objective is convex

$$\begin{aligned}\nabla J(\hat{\theta}) = 0 &\iff X^T X \hat{\theta} = X^T y, \\ H = \nabla^2 J &= \frac{1}{n} X^T X, \quad v^T H v = \frac{1}{n} \|Xv\|^2 \geq 0, \\ J(\hat{\theta} + v) - J(\hat{\theta}) &= \frac{1}{2n} \|Xv\|^2 \geq 0.\end{aligned}$$

- Every solution of the normal equations is a global minimizer; rank deficiency may leave multiple solutions.
- A minimizer exists: project y onto the closed finite-dimensional space $\text{Col}(X)$ (column space of the design matrix X).

When is the analytic solution unique?

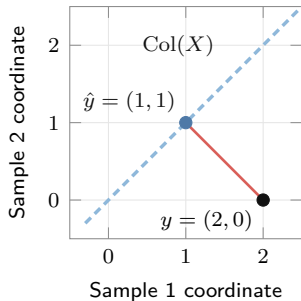
An inverse is justified only after the rank condition is checked

$$\begin{aligned}\text{rank}(X) = p &\iff Xv \neq 0 \text{ for every } v \neq 0 \iff X^T X \succ 0, \\ X^T X \hat{\theta} &= X^T y \implies \hat{\theta} = (X^T X)^{-1} X^T y.\end{aligned}$$

- Full column rank requires $n \geq p$, but $n \geq p$ alone does not guarantee it.
- In code, solve the least-squares problem with QR or SVD; the displayed inverse is an analytic expression.
- A full-rank matrix can still be nearly singular and yield unstable coefficients.

Least squares as a projection

Find the representable prediction $X\theta$ closest to y



The projection of black point y lies in $\text{Col}(X)$.

$$\hat{y} = P_X y, \quad X^T(y - \hat{y}) = 0.$$

The red residual is orthogonal to every feature column.

Two-dimensional example: $X = [1, 1]^T$, $y = [2, 0]^T$, $\hat{y} = [1, 1]^T$.

Derive the projection matrix

The fitted vector is a linear transformation of the observed responses

$$\begin{aligned}\hat{y} &= X\hat{\theta} = Py, & P &= X(X^\top X)^{-1}X^\top, \\ P^\top &= P, \\ P^2 &= X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top = P, \\ PX &= X, & X^\top(I - P) &= 0.\end{aligned}$$

- Assume full column rank here. Symmetry and idempotence identify an orthogonal projection.
- P preserves vectors in $\text{Col}(X)$ and removes their orthogonal component.

A Pythagorean proof of the best fit

Every competing prediction adds a perpendicular component of error

$$\begin{aligned}e &= y - \hat{y}, & \hat{y} &= Py, & z &\in \text{Col}(X), \\y - z &= e + (\hat{y} - z), & e^\top (\hat{y} - z) &= 0, \\ \|y - z\|^2 &= \|e\|^2 + \|\hat{y} - z\|^2 \geq \|e\|^2, \\ \mathbf{1} \in \text{Col}(X) &\implies \mathbf{1}^\top e = 0 \implies \bar{\hat{y}} = \bar{y}.\end{aligned}$$

- Equality holds only when the competing fitted vector is $z = \hat{y}$.
- An included intercept makes residuals sum to zero; it does not make residuals independent of inputs.

III-conditioning and remedies

Unique predictions, unique parameters?

Can the data distinguish different parameter values?

- **Full column rank.** $X^T X$ is positive definite; the OLS parameters are unique.
- **Rank deficiency.** There is a $v \neq 0$ such that $Xv = 0$.
- **Indistinguishable.** θ and $\theta + v$ make identical training predictions.
- **Choosing a solution.** The pseudoinverse selects the minimum Euclidean norm.

More features do not automatically add information. Predictions at new inputs may still differ.

The compact singular value decomposition

SVD decomposition:

$$X = U_r \Sigma_r V_r^T, \quad r = \text{rank}(X), \quad \Sigma_r = \text{diag}(s_1, \dots, s_r),$$
$$X \in \mathbb{R}^{n \times p}, \quad U_r \in \mathbb{R}^{n \times r}, \quad V_r \in \mathbb{R}^{p \times r}, \quad s_1 \geq \dots \geq s_r > 0.$$

- $U_r^T U_r = V_r^T V_r = I_r$: the columns are orthonormal.
- A parameter direction v_j produces the prediction direction $X v_j = s_j u_j$.
- Small s_j : a large coefficient change can produce a small change in the observed predictions.

Least squares in singular coordinates

Write $\theta = V_r a + v_0$, where $Xv_0 = 0$, and let $c_j = u_j^\top y$.

$$\|X\theta - y\|_2^2 = \sum_{j=1}^r (s_j a_j - c_j)^2 + \|(I - U_r U_r^\top)y\|_2^2.$$

$$\frac{\partial}{\partial a_j} (s_j a_j - c_j)^2 = 2s_j (s_j a_j - c_j) = 0 \implies \hat{a}_j = \frac{c_j}{s_j}.$$

$$\hat{\theta}_{\min} = \sum_{j=1}^r v_j \cdot \frac{u_j^\top y}{s_j} = \underbrace{V_r \Sigma_r^{-1} U_r^\top}_{X^+} y$$

$$X^+ = (X^\top X)^{-1} X^\top \text{ when } (X^\top X) \text{ is invertible}$$

Rank deficiency and minimum norm

Rank deficiency creates freedom in the coefficients, not in the fitted values

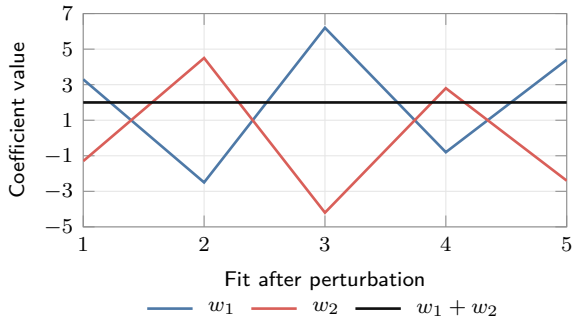
$$\arg \min_{\theta} \|X\theta - y\|_2^2 = \{X^+y + v_0 : v_0 \in \ker(X)\}.$$

$$\|X^+y + v_0\|_2^2 = \|X^+y\|_2^2 + \|v_0\|_2^2, \quad \hat{y} = XX^+y.$$

- X^+y is orthogonal to $\ker(X)$, so $v_0 = 0$ uniquely minimizes the parameter norm.
- Every least-squares minimizer has the same training prediction $\hat{y}$.
- Example: $X = [1 \ 1]$, $y = 1$. All $(a, 1 - a)^T$ fit exactly; the minimum-norm choice is $(1/2, 1/2)^T$.

Collinearity destabilizes coefficients

Similar predictions can conceal large coefficient changes



- When $x_2 \approx x_1$, the data mainly constrain $w_1 + w_2$.
- Small singular-value directions amplify data noise.
- Standardization cannot remove genuine collinearity.

Full rank does not imply stability

Existence and uniqueness are different from sensitivity to perturbations

$$\text{rank}(X) = p \iff s_p > 0, \quad \kappa_2(X) = \frac{s_1}{s_p}.$$

$$\Delta \hat{\theta} = X^+ \Delta y, \quad \|\Delta \hat{\theta}\|_2 \leq \frac{\|\Delta y\|_2}{s_p} \quad \text{when } X \text{ is fixed.}$$

- Full rank answers whether the coefficients are uniquely determined in exact arithmetic.
- A large condition number signals very different strengths across parameter directions.
- Whether sensitivity is unacceptable depends on noise, units, arithmetic precision, and the intended use. There is no universal cutoff.

A controlled near-collinear design

Two columns become nearly equal while the matrix remains full rank

Let q_1, q_2 be orthonormal. For the concrete two-sample example, use $q_1 = (1, 0)^\top$ and $q_2 = (0, 1)^\top$.

$$X_\epsilon = [q_1 + \epsilon q_2 \quad q_1 - \epsilon q_2] = \begin{bmatrix} 1 & 1 \\ \epsilon & -\epsilon \end{bmatrix}, \quad 0 < \epsilon \leq 1.$$

$$v_+ = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_- = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad X_\epsilon v_+ = \sqrt{2} q_1, \quad X_\epsilon v_- = \sqrt{2} \epsilon q_2.$$

$$s_+ = \sqrt{2}, \quad s_- = \sqrt{2} \epsilon, \quad \kappa_2(X_\epsilon) = \epsilon^{-1}.$$

A tiny response change

Keep the design fixed and perturb only its weak response direction

$$\begin{aligned}y &= q_1, & y' &= q_1 + \delta q_2, & X_\epsilon \hat{\theta}' &= y', \\ \hat{\theta}'_1 + \hat{\theta}'_2 &= 1, & \epsilon(\hat{\theta}'_1 - \hat{\theta}'_2) &= \delta.\end{aligned}$$

A tiny response change

Keep the design fixed and perturb only its weak response direction

$$y = q_1, \quad y' = q_1 + \delta q_2, \quad X_\epsilon \hat{\theta}' = y',$$

$$\hat{\theta}'_1 + \hat{\theta}'_2 = 1, \quad \epsilon(\hat{\theta}'_1 - \hat{\theta}'_2) = \delta.$$

$$\hat{\theta}' = \begin{bmatrix} 1/2 + \delta/(2\epsilon) \\ 1/2 - \delta/(2\epsilon) \end{bmatrix}, \quad \|\Delta \hat{\theta}\|_2 = \frac{|\delta|}{\sqrt{2}\epsilon}.$$

A tiny response change

Keep the design fixed and perturb only its weak response direction

$$y = q_1, \quad y' = q_1 + \delta q_2, \quad X_\epsilon \hat{\theta}' = y',$$

$$\hat{\theta}'_1 + \hat{\theta}'_2 = 1, \quad \epsilon(\hat{\theta}'_1 - \hat{\theta}'_2) = \delta.$$

$$\hat{\theta}' = \begin{bmatrix} 1/2 + \delta/(2\epsilon) \\ 1/2 - \delta/(2\epsilon) \end{bmatrix}, \quad \|\Delta\hat{\theta}\|_2 = \frac{|\delta|}{\sqrt{2}\epsilon}.$$

With $\epsilon = 10^{-4}$ and $\delta = 10^{-3}$:

$$\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \longrightarrow \begin{bmatrix} 5.5 \\ -4.5 \end{bmatrix}, \quad \|\Delta y\|_2 = 0.001, \quad \|\Delta\hat{\theta}\|_2 \approx 7.071.$$

Which predictions remain stable?

Coefficient error, training prediction error, and future prediction error differ

$$\Delta \hat{y} = XX^+ \Delta y, \quad \|XX^+\|_{\text{op}} \leq 1, \quad \|\Delta \hat{y}\|_2 \leq \|\Delta y\|_2.$$

$$\Delta f(x_*) = x_*^\top \Delta \hat{\theta} = \sum_{j=1}^r \frac{(x_*^\top v_j)(u_j^\top \Delta y)}{s_j}.$$

For the controlled example, with $\epsilon = 10^{-4}$ and $\delta = 10^{-3}$:

Quantity	Change	Reason
Training fitted vector	δq_2	Norm 0.001
Future input $x_* = (1, 1)^\top$	0	Uses the coefficient sum
Future input $x_* = (1, -1)^\top$	$\delta/\epsilon = 10$	Amplifies the weak direction

Stable predictions on the observed design do not guarantee stability for a new feature pattern.

Sampling variance in weak directions

The same singular-value division appears in statistical uncertainty

Fixed X with $\text{rank}(X) = p$: $y = X\theta_* + \xi$, $\mathbb{E}[\xi | X] = 0$, and $\text{Cov}(\xi | X) = \sigma^2 I$.

$$\hat{\theta} - \theta_* = \sum_{j=1}^p \frac{u_j^\top \xi}{s_j} v_j, \quad \text{Cov}(\hat{\theta} | X) = \sigma^2 V \text{diag}(s_j^{-2}) V^\top.$$

$$\text{Var}(v_j^\top \hat{\theta} | X) = \frac{\sigma^2}{s_j^2}.$$

- Tenfold smaller s_j gives hundredfold larger variance;

Numerical rank is a tolerance decision

A tiny positive singular value need not contain usable information

$$\mathcal{I}_\tau = \{j : s_j > \tau s_1\}, \quad \hat{\theta}_\tau = \sum_{j \in \mathcal{I}_\tau} \frac{u_j^\top y}{s_j} v_j, \quad \hat{y}_\tau = \sum_{j \in \mathcal{I}_\tau} u_j u_j^\top y.$$

- Truncated SVD keeps selected singular directions and removes the rest.
- A roundoff heuristic is $\tau \sim \max(n, p)u$, where u is unit roundoff. Measurement noise may justify stronger truncation.
- Try-and-error in order: inspect feature scales and singular values; check residuals and solver sensitivity; test the intended future prediction task.

Gradient descent for linear regression

Why is Gradient Descent (GD) not the closed-form solution X^+y ? The dimension p can be large, and the data may arrive in a stream.

Quadratic curvature determines step size and convergence

$$\theta_{t+1} = \theta_t - \frac{\eta}{n} X^T (X\theta_t - y),$$
$$H = \frac{1}{n} X^T X, \quad L = \lambda_{\max}(H), \quad 0 < \eta < \frac{2}{L}.$$

- With full column rank, this step size contracts error in every eigendirection.
- The largest curvature limits the stable step size.
- The convergence factor is $\max_j |1 - \eta\lambda_j|$.

Derivation: $\theta_{t+1} - \theta^* = (I - \eta H)(\theta_t - \theta^*)$.

Ill-conditioning slows gradient descent

A stable step size can still produce very slow progress

$$\begin{aligned}e_{t+1} &= (I - \eta H)e_t, & H &= \frac{1}{n}X^\top X, & e_t &= \theta_t - \hat{\theta}, \\ \eta_* &= \frac{2}{L + \mu}, & q_* &= \max_j |1 - \eta_* \lambda_j(H)| = \frac{\kappa_H - 1}{\kappa_H + 1}, \\ \kappa_H &= \frac{L}{\mu} = \kappa_2(X)^2, & \|e_t\|_2 &\leq q_*^t \|e_0\|_2.\end{aligned}$$

- Assume full column rank, with $0 < \mu = \lambda_{\min}(H) \leq L$.
- If $\kappa_2(X) = 10^3$, then $\kappa_H = 10^6$ and $q_* \approx 0.999998$.
- Directional rescaling can help improve the efficiency.

Choose the right remedy

A reliable computation cannot create information that the data do not contain

Problem	Response	What changes?
Inconsistent units	Center and scale	Coordinates; OLS fitted space is preserved
Avoidable roundoff	QR or SVD solver	Computation of the same LSQ objective
Weak input directions	Better measurements or a redesigned sample	Information in the data
Unreliable directions	Truncation, Ridge, or Lasso	Model or objective; bias may increase

Regularization: Ridge and Lasso

Ridge Regression

Constrain parameter directions that the data determine poorly

$$\begin{aligned} J(\theta; \lambda) &= \frac{\|X\theta - y\|^2}{2n} + \frac{\lambda}{2} \|w\|^2, \\ D &= \text{diag}(0, 1, \dots, 1), \\ \hat{\theta} &= (X^\top X + n\lambda D)^{-1} X^\top y. \end{aligned}$$

- $\lambda > 0$ shrinks slopes; the intercept is unpenalized.
- Small singular-value directions receive stronger relative shrinkage.
- Select λ using validation data.

Ridge filters singular directions

Derive the exact shrinkage factors

$$\min_w \frac{\|X\theta - y\|_2^2}{2n} + \frac{\lambda}{2} \|\theta\|_2^2, \quad X = U_r \operatorname{diag}(s_j) V_r^\top.$$
$$\hat{\theta}_\lambda = \sum_{j=1}^r \frac{s_j}{s_j^2 + n\lambda} (u_j^\top y_c) v_j, \quad X \hat{\theta}_\lambda = \sum_{j=1}^r \underbrace{\frac{s_j^2}{s_j^2 + n\lambda}}_{h_j} (u_j^\top y_c) u_j.$$

Model $\theta = \sum_j a_j v_j$. The scalar normal equation is $(s_j^2 + n\lambda)a_j = s_j u_j^\top y_c$. For $\lambda > 0$, null-space coefficients are zero.

Ridge stabilizes the controlled example

Compare stability and training fit under one controlled perturbation

Return to the **intercept-free** toy, with $n = 2$, $\epsilon = 10^{-4}$, and $y = q_1 + 10^{-3}q_2$.

$$\hat{\theta}_{1,\lambda} = \frac{1}{2(1+\lambda)} + \frac{10^{-3}\epsilon}{2(\epsilon^2 + \lambda)}, \quad \hat{\theta}_{2,\lambda} = \frac{1}{2(1+\lambda)} - \frac{10^{-3}\epsilon}{2(\epsilon^2 + \lambda)}.$$

λ	$\hat{\theta}_1$	$\hat{\theta}_2$	Training MSE
0	5.500000	-4.500000	0
10^{-6}	0.549504	0.450495	4.9015×10^{-7}
10^{-4}	0.500450	0.499450	5.0490×10^{-7}

At $\lambda = 10^{-4}$: strong direction $h_+ \approx 0.9999$; weak direction $h_- \approx 10^{-4}$.

Lasso Regression

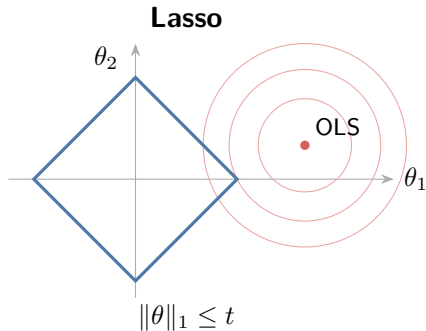
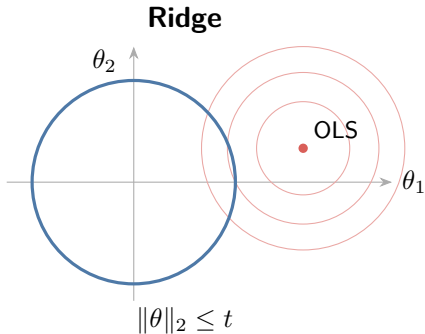
Ridge shrinks coefficients; an absolute-value penalty can set them to zero

$$\min_{\theta} \frac{\|y - X\theta\|_2^2}{2n} + \lambda \|\theta\|_1, \quad \|\theta\|_1 = \sum_{j=1}^d |\theta_j|.$$

- Assume $n > 0$, $\lambda \geq 0$; the intercept is unpenalized.
- Fit centering and scaling on the training fold only.

Why do corners encourage zeros?

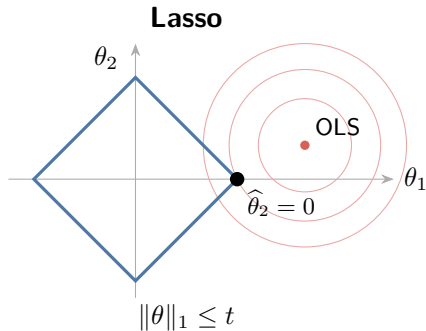
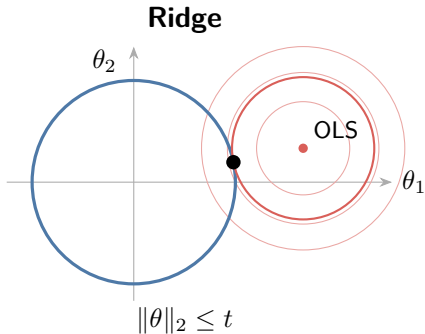
Where does the smallest loss contour first touch the constraint?



Constructed contours; both constraint radii use $t = 1.2$.

Why do corners encourage zeros?

Where does the smallest loss contour first touch the constraint?



Constructed contours; both constraint radii use $t = 1.2$.

The L_1 boundary has axis-aligned corners; a contact at a corner gives a zero coefficient.

Optimality at a zero coefficient

At a kink, optimality uses a set of possible slopes

$$\partial|u| = \begin{cases} \{1\}, & u > 0, \\ [-1, 1], & u = 0, \\ \{-1\}, & u < 0, \end{cases} \quad 0 \in -\frac{X_j^\top(y - X\hat{\theta})}{n} + \lambda\partial|\hat{\theta}_j|.$$

$$\hat{\theta}_j = 0 \implies \left| \frac{X_j^\top(y - X\hat{\theta})}{n} \right| \leq \lambda.$$

- A nonzero coefficient requires residual correlation $= \lambda \operatorname{sign}(\hat{\theta}_j)$.
- Convexity makes the full set of these conditions sufficient for a global minimum.

Deriving a coordinate-descent update

Hold the other coefficients fixed; subtract their current predictions

$$r_{-j} = y - \sum_{k \neq j} X_k \theta_k, \quad a_j = \frac{\|X_j\|_2^2}{n} > 0, \quad c_j = \frac{X_j^\top r_{-j}}{n}.$$
$$\frac{\|r_{-j} - X_j \theta_j\|_2^2}{2n} + \lambda |\theta_j| = \frac{a_j}{2} \theta_j^2 - c_j \theta_j + \lambda |\theta_j| + \text{constant},$$

- With our standardization, $a_j = 1$; cycle through all retained columns.
- Use warm starts along a decreasing λ sequence; stop using an optimality tolerance.

Ridge and Lasso trace different paths

Constructed normalized orthogonal design: $c = (2, 0.8, -0.35)^T$

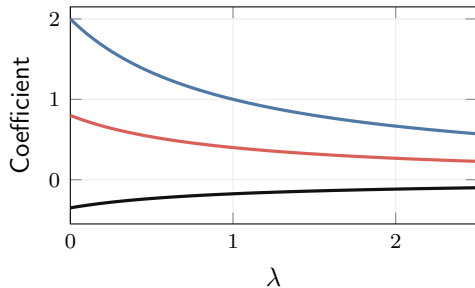
Increase λ from zero.

Which coefficients reach exactly zero?

Ridge and Lasso trace different paths

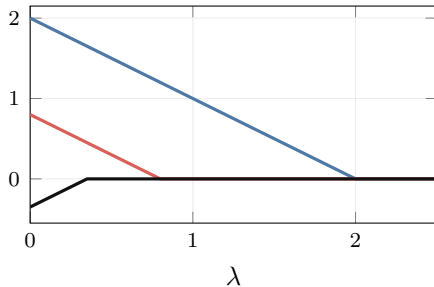
Constructed normalized orthogonal design: $c = (2, 0.8, -0.35)^T$

Ridge



w_1 w_2 w_3

Lasso



Ridge: $\hat{\theta}_j = c_j / (1 + \lambda)$. Lasso: $\hat{\theta}_j$: no closed form solution.

Choose a penalty for the modeling task

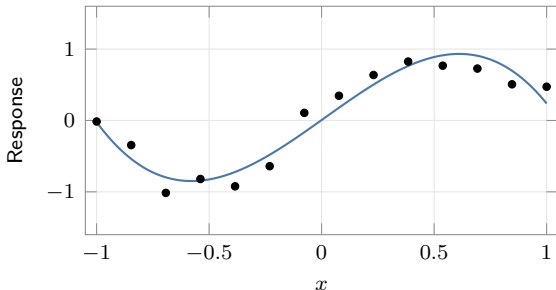
What tradeoff are we willing to make?

	Ridge	Lasso
Penalty	$\lambda \ \theta\ _2^2 / 2$	$\lambda \ \theta\ _1$
Typical behavior	Continuous shrinkage	Shrinkage with possible exact zeros
Useful preference	Many small effects; stable prediction	A smaller feature set
Correlated inputs	Spreads weight across similar columns	Selection can be unstable
Tuning	Validation / cross-validation	Validation / cross-validation

Feature expansion: polynomials and kernels

Can a linear model bend?

5. Feature expansion



Same fitting machinery. Richer inputs.

$$\hat{y} = b + w_1x + w_2x^2 + w_3x^3$$

$$\phi(x) = [x \quad x^2 \quad x^3]^T$$

$$\hat{y} = b + w^T \phi(x)$$

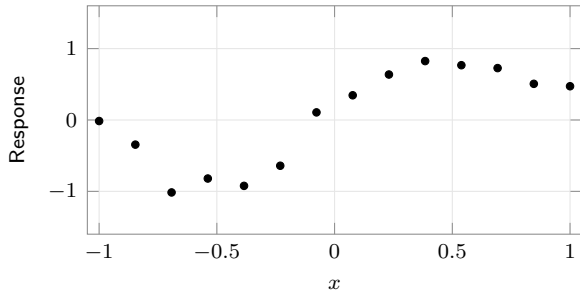
Linear in w .

Curved in x .

Feature design changes the functions we can fit.

How much curvature is useful?

Fourteen training observations. Which degree would you choose?



Degree 1?

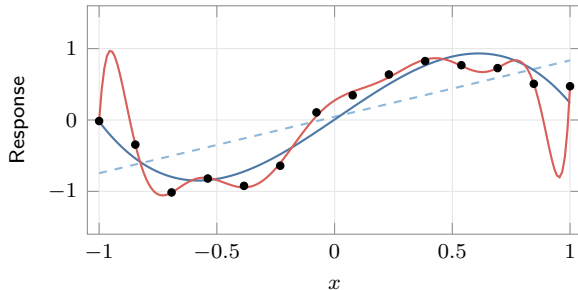
Degree 3?

Degree 12?

Make a prediction first.

How much curvature is useful?

Fourteen training observations. Which degree would you choose?



Dashed: degree 1

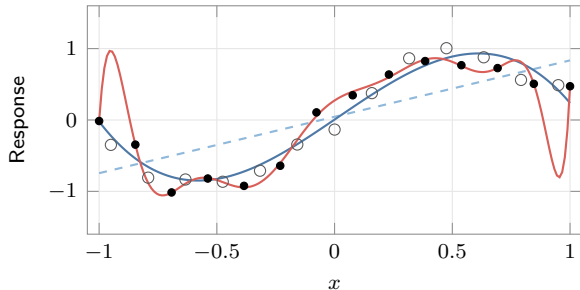
Blue: degree 3

Red: degree 12

Which one wins on new data?

How much curvature is useful?

Fourteen training observations. Which degree would you choose?



Separate validation observations

Degree	Train MSE	Val. MSE
1	0.1793	0.1455
3	0.0300	0.0149
12	0.0007	0.2897

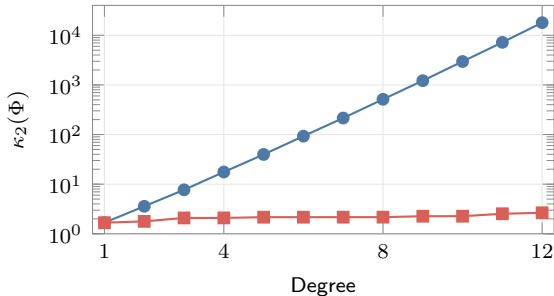
● Training ○ Validation

Better fit $\neq$ better prediction.

Choose degree and penalty using validation; keep a final test set untouched.

Expansion can bring back ill-conditioning

Same polynomial space. Different coordinates.



Monomials

$$1, x, x^2, \dots, x^q$$

Chebyshev basis

$$T_0(x), T_1(x), \dots, T_q(x)$$

At degree 12:

$$\kappa_2 : 18,000 \longrightarrow 2.65$$

Scale the input; solve with QR or SVD.

A basis change preserves the OLS model space, but generally changes an isotropic ridge penalty.

A dot product can hide six features

Two-dimensional inputs: $x = (x_1, x_2)$ and $z = (z_1, z_2)$.

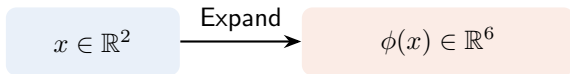
$$k(x, z) = (1 + x^T z)^2$$

Can this be a dot product in another space?

A dot product can hide six features

Two-dimensional inputs: $x = (x_1, x_2)$ and $z = (z_1, z_2)$.

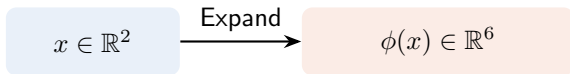
$$\phi(x) = [1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, \sqrt{2}x_1x_2, x_2^2]^T$$



A dot product can hide six features

Two-dimensional inputs: $x = (x_1, x_2)$ and $z = (z_1, z_2)$.

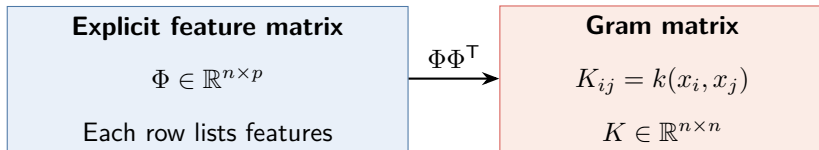
$$\phi(x) = [1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, \sqrt{2}x_1x_2, x_2^2]^\top$$



$$\phi(x)^\top \phi(z) = 1 + 2x^\top z + (x^\top z)^2 = (1 + x^\top z)^2.$$

Compare points without listing features

The kernel trick: compute $k(x, z) = \langle \phi(x), \phi(z) \rangle$ directly.



100 inputs, degree at most 3

$p = \binom{103}{3} = 176,851$ distinct terms.

But the sample size still matters

Dense K : $O(n^2)$ storage; direct solve: $O(n^3)$.

How to solve the kernelized linear regression?

$$\min_{\theta} J(\theta) = \frac{1}{2n} \sum_{i=1}^n (\phi_i^{\top} \theta - y)^2 + \frac{1}{2} \cdot \lambda \|\theta\|_2$$

How to solve the kernelized linear regression?

$$\min_{\theta} J(\theta) = \frac{1}{2n} \sum_{i=1}^n (\phi_i^{\top} \theta - y)^2 + \frac{1}{2} \cdot \lambda \|\theta\|_2$$

$$\theta^* = (\Phi^{\top} \Phi + \lambda I)^{-1} \Phi^{\top} y$$

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ϕ_i can have a high dimension d . (what is the size of $\Phi^{\top} \Phi$?)

How to solve the kernelized linear regression?

$$\min_{\theta} J(\theta) = \frac{1}{2n} \sum_{i=1}^n (\phi_i^{\top} \theta - y)^2 + \frac{1}{2} \cdot \lambda \|\theta\|_2^2$$

$$\theta^* = (\Phi^{\top} \Phi + \lambda I)^{-1} \Phi^{\top} y$$

ϕ_i can have a high dimension d . (what is the size of $\Phi^{\top} \Phi$?)

$$\theta^* = (\Phi^{\top} \Phi + \lambda I_d)^{-1} \Phi^{\top} y = \Phi^{\top} (\Phi \Phi^{\top} + \lambda I_n) y = \Phi^{\top} (K + \lambda I_n) y$$

How to solve the kernelized linear regression?

For new x , predict

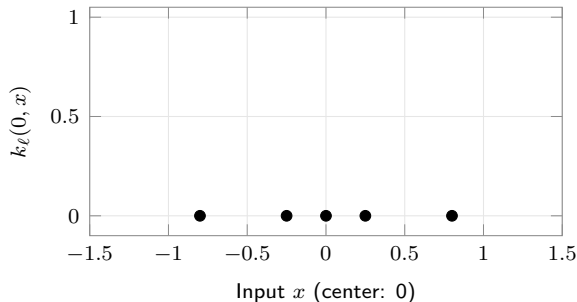
$$\phi(x)^\top \theta^* = \phi(x)^\top \Phi^\top (K + \lambda I_n) y = v^\top (K + \lambda I_n) y$$

,

where $v^\top = [K(x, x_1), K(x, x_2), \dots, K(x, x_n)]$.

How local is an RBF similarity?

Compare each input to $x = 0$. Before fitting any regression coefficients.



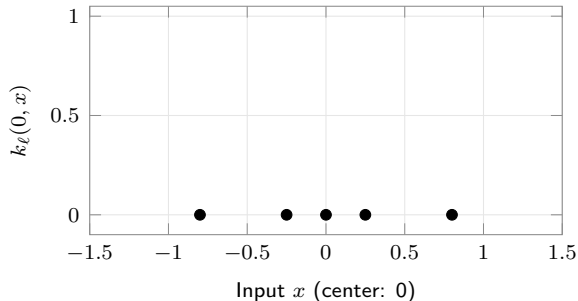
Which point is more similar?

$x = 0.25$ or $x = 0.8$

What should “nearby” mean?

How local is an RBF similarity?

Compare each input to $x = 0$. Before fitting any regression coefficients.



$$k_\ell(x, z) = e^{-\|x-z\|^2/(2\ell^2)}$$

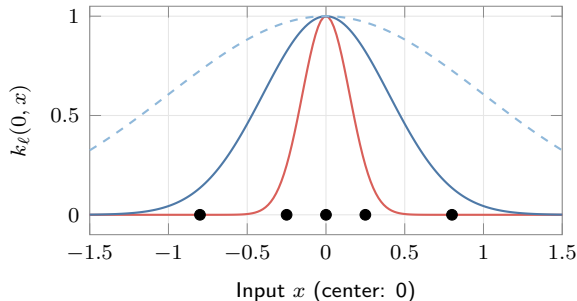
At distance ℓ :

$$k = e^{-1/2} \approx 0.607$$

Predict the effect of increasing ℓ .

How local is an RBF similarity?

Compare each input to $x = 0$. Before fitting any regression coefficients.



$\ell = 0.15$: **narrow**

$\ell = 0.40$: **intermediate**

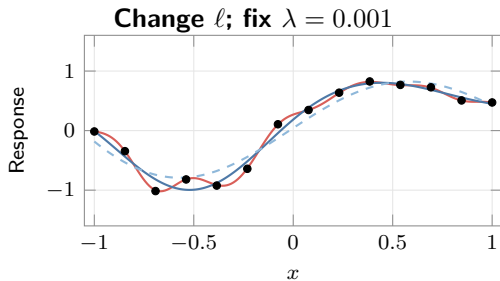
$\ell = 1.00$: **broad**

Scale inputs before measuring distance.

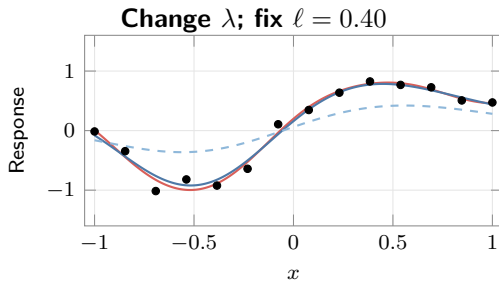
The bandwidth sets similarity range; learned coefficients set the prediction.

Bandwidth and penalty do different jobs

Same observations. Centered RBF kernel ridge with an unpenalized intercept.



$\ell = .12$ $\ell = .40$ $\ell = 1.00$

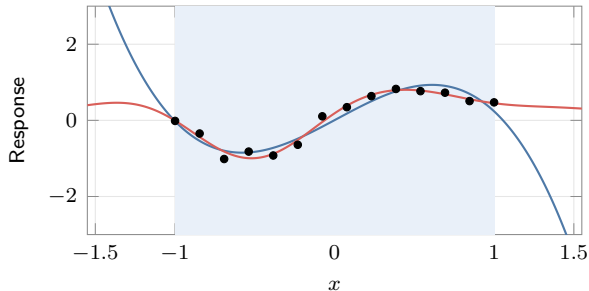


$\lambda = .0001$ $\lambda = .01$ $\lambda = .30$

Tune bandwidth and penalty together on validation data. They are not interchangeable.

Beyond the observed range?

Both methods fit inside the blue window. What supports a prediction outside it?



Cubic OLS

Powers continue globally.

RBF kernel ridge

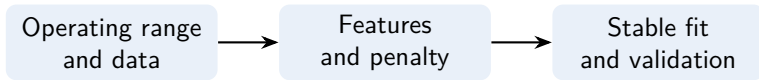
Local similarities fade.

Neither curve supplies evidence about an unseen regime.

Match validation to the prediction domain. Feature expansion preserves the need for judgment.

Return to tomorrow's electricity forecast

Choose the model, solver, penalty, and representation for the prediction task



- Can you derive the LSQ solution and state its rank condition?
- Can you distinguish numerical error, sensitivity, and overfitting?
- What evidence justifies Ridge, Lasso, polynomial features, or a kernel?

Begin with an interpretable benchmark; add complexity when validation supports it.

Statistical assumptions

$$y = X\theta^* + \varepsilon, \quad X \in \mathbb{R}^{n \times p}, \quad \text{rank}(X) = p,$$
$$\mathbb{E}[\varepsilon \mid X] = 0, \quad \text{Cov}(\varepsilon \mid X) = \sigma^2 I_n.$$

- The conditional mean is linear in the chosen features.
- p counts every fitted coefficient, including an intercept.
- For now, condition on X and average over training noise.
- Gaussian noise is unnecessary for the expectation identities.

Full column rank gives $n \geq p$. Equal noise variance and zero conditional mean are substantive assumptions.

Which error are we averaging?

Quantity	Squared error for one fitted model
Training error	$\widehat{R}_{\text{train}} = \frac{1}{n} \ y - X\widehat{\theta}\ ^2$
Parameter error	$\ \widehat{\theta} - \theta^*\ ^2$
Mean prediction error at X	$\mathcal{E}_X = \frac{1}{n} \ X\widehat{\theta} - X\theta^*\ ^2$
Fresh-response error at X	$\frac{1}{n} \ y' - X\widehat{\theta}\ ^2$, with independent new noise

Predicting the true mean and predicting a noisy response differ by irreducible noise.

OLS estimation error

Substitute the data-generating model into the closed-form solution

$$\begin{aligned}\hat{\theta} &= (X^T X)^{-1} X^T y \\ &= (X^T X)^{-1} X^T (X\theta^* + \varepsilon) \\ &= \theta^* + (X^T X)^{-1} X^T \varepsilon.\end{aligned}$$

The OLS estimator is unbiased: $\mathbb{E}[\hat{\theta}] = \theta^*$.

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$$\boxed{\hat{\theta} - \theta^* = A\varepsilon}, \quad A = (X^T X)^{-1} X^T \in \mathbb{R}^{p \times n}.$$

The OLS estimator is unbiased: $\mathbb{E}[\hat{\theta}] = \theta^*$.

Unbiasedness and covariance

$$\mathbb{E}[\hat{\theta} \mid X] = \theta^* + A\mathbb{E}[\varepsilon \mid X] = \theta^*,$$

$$\begin{aligned}\text{Cov}(\hat{\theta} \mid X) &= A \text{Cov}(\varepsilon \mid X) A^T \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \\ &= \boxed{\sigma^2 (X^T X)^{-1}}.\end{aligned}$$

OLS is centered at the true parameter. Its variance can still be large.

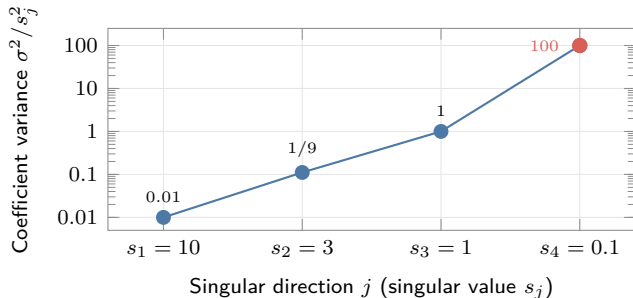
Expected squared parameter error

For any random vector z , $\mathbb{E}\|z\|^2 = \|\mathbb{E}z\|^2 + \text{tr}(\text{Cov}(z))$

$$\begin{aligned}\mathbb{E}[\|\hat{\theta} - \theta^*\|^2 \mid X] &= \underbrace{\|\mathbb{E}[\hat{\theta} \mid X] - \theta^*\|^2}_0 + \text{tr}(\text{Cov}(\hat{\theta} \mid X)) \\ &= \sigma^2 \text{tr}((X^\top X)^{-1}) \\ &= \boxed{\sigma^2 \sum_{j=1}^p \frac{1}{s_j^2}}, \quad s_j > 0.\end{aligned}$$

Here s_j are the singular values of X .

Ill-conditioning and statistical error



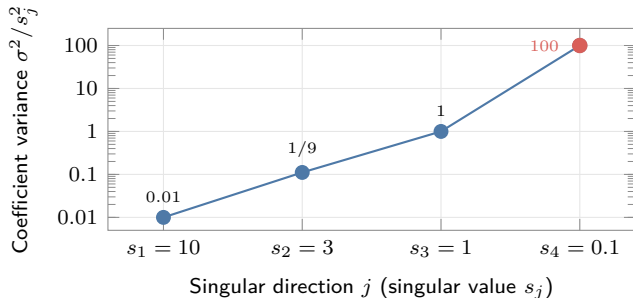
For $X = USV^T$,

$$\text{Var}(v_j^T \hat{\theta} \mid X) = \frac{\sigma^2}{s_j^2}.$$

Which direction dominates?

QR and SVD improve numerical computation. Weakly observed directions still have statistical uncertainty.

Ill-conditioning and statistical error



For $X = USV^T$,

$$\text{Var}(v_j^T \hat{\theta} \mid X) = \frac{\sigma^2}{s_j^2}.$$

Which direction dominates?

With $\sigma^2 = 1$,

$$\mathbb{E} \|\hat{\theta} - \theta^*\|^2 \approx 101.12.$$

The weakest direction contributes 100.

QR and SVD improve numerical computation. Weakly observed directions still have statistical uncertainty.

Prediction error at one input

Let $x_0 \in \mathbb{R}^p$ be a fixed feature vector, including an intercept if used

$$\begin{aligned}\mathbb{E}[(x_0^\top \hat{\theta} - x_0^\top \theta^\star)^2 \mid X] &= x_0^\top \text{Cov}(\hat{\theta} \mid X) x_0 \\ &= \boxed{\sigma^2 x_0^\top (X^\top X)^{-1} x_0}.\end{aligned}$$

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For a fresh response $y_0 = x_0^\top \theta^\star + \varepsilon_0$,

$$\mathbb{E}[(y_0 - x_0^\top \hat{\theta})^2 \mid X] = \underbrace{\sigma^2}_{\text{new response noise}} + \underbrace{\sigma^2 x_0^\top (X^\top X)^{-1} x_0}_{\text{estimation error}}.$$

Prediction at new random inputs

The test pair (x_0, ε_0) is independent of the training sample. Let $Q = \mathbb{E}[x_0 x_0^\top]$

$$R(\theta) = \mathbb{E}_{x_0, \varepsilon_0}[(y_0 - x_0^\top \theta)^2],$$

$$y_0 = x_0^\top \theta^* + \varepsilon_0, \quad \mathbb{E}[\varepsilon_0 \mid x_0] = 0, \quad \mathbb{E}[\varepsilon_0^2 \mid x_0] = \sigma^2,$$

$$R(\theta) - R(\theta^*) = (\theta - \theta^*)^\top Q (\theta - \theta^*), \quad R(\theta^*) = \sigma^2.$$

For a realized $\hat{\theta}$, $R(\hat{\theta})$ averages over a new test example.

Q is a second-moment matrix. It equals a covariance matrix only when the feature mean is zero.

Expected excess prediction risk

Average the population risk over the noise that produced $\hat{\theta}$

$$\begin{aligned}\mathbb{E}[R(\hat{\theta}) - R(\theta^*) \mid X] &= \mathbb{E}[(\hat{\theta} - \theta^*)^\top Q(\hat{\theta} - \theta^*) \mid X] \\ &= \text{tr}\left(Q \text{Cov}(\hat{\theta} \mid X)\right) \\ &= \boxed{\sigma^2 \text{tr}(Q(X^\top X)^{-1})}.\end{aligned}$$

Error depends on how the test-input directions align with the directions learned from the training design.

Expected excess prediction risk

Average the population risk over the noise that produced $\hat{\theta}$

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If test inputs are uniform over the rows of X , their conditional second moment is $X^\top X/n$, recovering $\sigma^2 p/n$.

Error depends on how the test-input directions align with the directions learned from the training design.

An expected generalization bound

A lower bound on the design eigenvalues gives a usable upper bound on risk

$$\hat{G} = \frac{1}{n} X^\top X, \quad \lambda_{\min}(\hat{G}) \geq a > 0,$$

$$\mathbb{E}[R(\hat{\theta}) - R(\theta^*) \mid X] = \frac{\sigma^2}{n} \operatorname{tr}(Q \hat{G}^{-1}) \leq \boxed{\frac{\sigma^2 \operatorname{tr}(Q)}{na}}.$$

If $Q = I_p$, the bound becomes $\sigma^2 p / (na)$.

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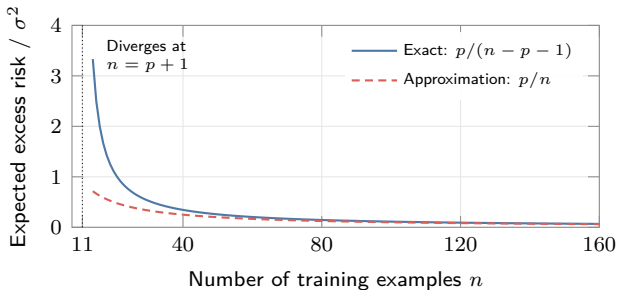
$$\mathbb{E}[R(\hat{\theta}) - R(\theta^*) \mid X] = \frac{\sigma^2}{n} \operatorname{tr}(Q \hat{G}^{-1}) \leq \boxed{\frac{\sigma^2 \operatorname{tr}(Q)}{na}}.$$

If $Q = I_p$, the bound becomes $\sigma^2 p / (na)$.

When does an $O(p/n)$ statement become informative?

When the design remains well supported: a stays away from zero.

Learning curve under Gaussian inputs



$p = 10$, Gaussian inputs, no intercept.

n	Excess risk/ σ^2
20	$10/9 \approx 1.111$
50	$10/39 \approx 0.256$
100	$10/89 \approx 0.112$

How many samples are enough?

Both p and the tolerated error matter.

Regression Evaluation Metrics

- $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
- $\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
- $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$
- $R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n-1}{n-p}$
- $\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$